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WIRELESS CONNECTION SYSTEM, INFORMATION TERMINAL, DISPLAY DEVICE, AND WIRELESS CONNECTION METHOD

Abstract

A wireless connection system executes wireless connection between an information terminal of a user and a display device installed in environment. In a case where an information terminal existing in the vicinity thereof is detected, the display device displays device ID information related to authentication for wireless connection with the detected information terminal (or an image obtained by coding it) on a display screen. The information terminal confirms whether the device ID information read from the display screen match device ID information of an own device or not, and establishes wireless connection using an authentication ID for wireless connection with the display device after matching is confirmed. The display device establishes wireless connection with the information terminal using an authentication ID.

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Background/Summary

[0001] This application is the Continuation of U.S. application Ser. No. 17/783,717, filed Jun. 9, 2022, which is the U.S. National Phase under 35 U.S.C. 371 of International Application No. PCT/JP2019/048538, filed Dec. 11, 2019, and the entire contents of each of which Applications are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present invention relates to a technique of a wireless connection system, an information terminal, and a display device.

BACKGROUND ART

[0003] Conventionally, wireless connection may be made between an information terminal such as a head mounted display (HMD), a wearable terminal, a smartphone, or a tablet terminal, which is worn or carried by a user, and a display device such as a tablet terminal or a television device, which is installed in the environment. In particular, in a case where the number of inexpensive display devices increases, it is supposed that a large number of display devices is installed in environment including stores, public spaces, vehicles, and the like. It is also supposed a scene that a plurality of information terminals of a plurality of unspecified users and a plurality of display devices make wireless connection in a multiple-to-multiple relationship in the environment. For example, display devices such as tablet terminals are respectively prepared on tables in a store such as a restaurant. A user who is a customer wirelessly connects his or her information terminal to a selected display device, and uses display of a menu, ordering of goods or services, payment, and other functions.

[0004] As an example of a conventional technique, Japanese Patent Application Laid-Open Publication No. 2010-136238 (Patent Document 1) can be cited. Patent Document 1 describes that a process of recognizing and storing communication devices as communication partners with each other is simplified for a user, communication devices that can greatly develop a cooperation function among the communication devices in a market and its usage are provided, and the following. This communication device stores identification information of a communication device capable of wireless communication; receives identification information from communication devices in a communicable area; presents a list of communication devices that can communicate at that time on the basis of the received identification information and the stored identification information when wireless communication is started; receives selection of a communication device from the presented list; and starts wireless communication with the selected communication device. Further, this communication device captures a bar code including identification information of a communication device, and obtains the identification information of the communication device from the bar code information.

RELATED ART DOCUMENTS

Patent Documents

[0005] Patent Document 1: Japanese Patent Application Laid-Open Publication No. 2010-136238

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

[0006] As described above, there is a need for each user to wirelessly connect his or her information terminal to a desired display device among a plurality of display devices and use a display function in the environment. In such a situation, in the multiple-to-multiple relationship

between a plurality of information terminals and a plurality of display devices, each information terminal and each display device need to make appropriate and secure wireless connection with a selected and authorized partner. Since it is the wireless connection, there are a case where a plurality of display devices exists within a range that radio waves reach from an information terminal of a user, and a case where a plurality of information terminals exists within a range that radio waves reach from a display device. Further, there are a risk that an information terminal of a user is wirelessly connected to a non-target display device that does not exist in front of the information terminal, and a risk that a display device is wirelessly connected to a non-target information terminal that does not exist in front of the display device. If unlimited wireless connection between multiple devices and multiple devices is allowed, there are problems in terms of security and resources. Therefore, it is necessary to limit and control wireless connection using authentication.

[0007] Conventionally, in wireless communication interfaces such as Bluetooth (registered trademark) and Wi-Fi (registered trademark), it is necessary to use authentication identification information (authentication ID) in order to make secure wireless connection. For example, in case of Bluetooth, a specified authentication ID called a passkey (or a passcode, etc.) is used. For example, in case of Wi-Fi, a specified authentication ID such as SSID and password is used. For the sake of explanation, the passkey and the like are collectively referred to as an authentication ID.

[0008] However, for example, in a case where an information terminal such as an HMD (in other words, a wireless device having a wireless communication function) is to be wirelessly connected to a display device, information input means provided in the information terminal is limited. Therefore, it may be troublesome or difficult for the user to input the authentication ID. For example, conventionally, a general HMD does not include an input device such as a keyboard or a touch panel.

[0009] It is an object of the present invention to provide a technique capable of reducing the time and effort of a user for an input operation of authentication information at the time of wireless connection, and realizing appropriate and secure wireless connection between a desired information terminal and a desired display device even in multiple-to-multiple environment with respect to a technique of wirelessly connecting an information terminal of the user to a display device.

Means for Solving the Problem

[0010] A representative embodiment of the present invention has a configuration described below. A wireless connection system according to one embodiment is a wireless connection system for executing wireless connection between an information terminal worn or carried by a user and a display device installed in environment. In this case, the display device is configured to: display, in a case where an information terminal existing in a vicinity thereof is detected, device ID information related to authentication for wireless connection with the detected information terminal or an image obtained by coding the device ID information on a display screen; and establish the wireless connection by using an authentication ID for the wireless connection with the information terminal. Further, the information terminal is configured to: read the device ID information or the image on the display screen of the display device; confirm whether the read device ID information matches device ID information of an own device or not; and establish, after the matching is confirmed, the wireless connection using the authentication ID for the wireless connection with the display device.

Effects of the Invention

[0011] According to the representative embodiment of the present invention, it is possible to reduce the time and effort of a user for an input operation of authentication information at the time of wireless connection, and realize appropriate and secure wireless connection between a desired information terminal and a desired display device in multiple-to-multiple environment with respect to a technique of wirelessly connecting an information terminal of the user to a display device.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0012] FIG. 1 is a view illustrating a configuration of a wireless connection system according to a first embodiment of the present invention;

[0013] FIG. 2 is a view illustrating an environment example and a procedure outline in a wireless connection method according to the first embodiment;

[0014] FIG. 3 is a view illustrating an example of an appearance configuration of an information terminal (HMD) according to the first embodiment;

[0015] FIG. 4 is a view illustrating an example of a functional block configuration of the information terminal according to the first embodiment;

[0016] FIG. 5 is a view illustrating an example of a functional block configuration of a display device according to the first embodiment;

[0017] FIG. 6 is a view illustrating a processing sequence between the information terminal and the display device according to the first embodiment;

[0018] FIG. 7 is a view illustrating an example of a device list screen according to the first embodiment;

[0019] FIG. 8 is a view illustrating an example of a code screen according to the first embodiment;

[0020] FIG. 9 is a view illustrating a processing sequence between the information terminal and the display device by the wireless connection system and the method according to the first embodiment of the present invention;

[0021] FIG. 10 is a view illustrating an example of a device list screen according to a second embodiment;

[0022] FIG. 11 is a view illustrating a display example by an information terminal (HMD) according to the second embodiment;

[0023] FIG. 12 is a view illustrating a processing sequence between an information terminal and a display device by a wireless connection system and a method thereof according to a third embodiment of the present invention;

[0024] FIG. 13 is a view illustrating an example of a device code list screen according to the third embodiment;

[0025] FIG. 14 is a view illustrating a processing sequence between an information terminal and a display device by a wireless connection system and a method thereof according to a fourth embodiment of the present invention; and

[0026] FIG. 15 is a view illustrating an example of a code screen according to the fourth embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0027] Hereinafter, embodiments of the present invention will be described in detail with reference to the drawings. Note that in all drawings, the same reference numerals are respectively in principle applied to the same portions, and repeated explanation thereof will be omitted.

First Embodiment

[0028] A wireless connection system and a method thereof according to a first embodiment of the present invention will be described with reference to FIG. 1 to FIG. 8.

[Wireless Connection System]

[0029] FIG. 1 illustrates a configuration of a wireless connection system according to the first embodiment. As illustrated on an upper side of FIG. 1, the wireless connection system according to the first embodiment includes a plurality (M) of information terminals 1 of a plurality (M) of users and a plurality (N) of display devices 2 in certain environment. This system is a multiple-to-multiple system, that is, a system in which wireless connection can be made between the plurality (M) of information terminals 1 and the plurality (N) of display devices 2. This wireless connection

system executes wireless connection between an information terminal **1** of a user and a display device **2** selected by the user with authentication.

[0030] In this wireless connection system, by basically combining one of a plurality (M) of unspecified devices with one of a plurality (N) of unspecified devices freely in the environment, wireless connection between one desired information terminal and one desired display device is permitted for each user. Authentication (including device check) in this wireless connection system is authentication for wirelessly connect one desired information terminal of the user to one display device appropriately and simply between the plurality of devices and the plurality of devices.

[0031] The plurality (M) of users is respectively indicated by users U1, U2, U3, . . . , and UM. The plurality (M) of information terminals **1** is respectively indicated by information terminals T1, T2, T3, . . . , and TM. The plurality (N) of display devices **2** is respectively indicated by display devices D1, D2, . . . , and DN. The number M of the information terminals **1** may be different from the number N of the display devices **2**, or they may be $M \geq 1$ and $N \geq 1$. Note that a one-to-multiple scene of one information terminal **1** and a plurality (N) of display devices **2** and a multiple-to-one scene of the plurality (M) of information terminals **1** and one display device **2** are also included within the applicable range of the present system and method.

[0032] For example, in the environment such as a store, the plurality (N) of display devices **2** that is permitted to be used by a user and the information terminal **1** that have succeeded in authentication is installed. The user wears or carries the information terminal **1** s such as an HMD **1A**, for example. Note that the information terminal **1** may temporarily be provided to the user in the environment such as the store.

[0033] This wireless connection system has a mechanism in which a desired selected display device **2** in front of a certain user among a plurality of display devices **2** can be connected wirelessly to an information terminal **1** of the user easily and surely in the environment. In this system, even in a case where the information terminal **1** is an HMD or the like, the time and effort required for the user to input an authentication ID is reduced, and a simple method that is highly convenient for the user can be provided. Further, in this system, even in a case where a large number of devices that are candidates for connection exist in the environment, it is possible to appropriately make wireless connection between the desired information terminal **1** and the desired display device **2**, wireless connection between unintended devices is prevented, and security is enhanced. After the wireless connection between the information terminal **1** and the display device **2** is established, the user can utilize a display function and the like of the display device **2**.

[0034] A lower side of FIG. **1** illustrates an outline of wireless connection between one information terminal **1** (for example, the HMD **1A**) of one user and one display device **2**. The information terminal **1** of the user typically corresponds to the HMD **1A** (including smart glasses and the like) or a smartphone **1B**, but is not limited to these. The display device **2** typically corresponds to a tablet terminal, but is not limited to this.

[0035] Each of the information terminal **1** and the display device **2** includes a wireless communication function. In particular, the wireless communication function is a wireless communication function such as a short-range or medium-range wireless communication function corresponding to the environment, and uses Bluetooth (registered trademark) as a wireless communication interface, for example. The display device **2** has a function of detecting nearby information terminals **1** in addition to the display function. For example, the display device **2** can detect the nearby information terminal **1** by detecting a beacon (radio signal) emitted by the information terminal **1** using the provided wireless communication function. Alternatively, the display device **2** may detect the nearby information terminals **1** by using a camera **22** and other sensors. Each of the information terminals **1** has a function of being able to read an image or information displayed on a display screen **20** of the display device **2**. For example, the information terminal **1** recognizes an image or information in the display screen **20** from an image obtained by capturing the display screen **20** using a provided camera **12**.

[0036] For example, a user looks at the display screen **20** of the display device **2** in a state where the user wears an HMD **1A**. In a case where an information terminal **1** that appears in the vicinity is detected, the display device **2** displays authentication information for wirelessly connecting with the detected information terminal **1** on the display screen **20**. The authentication information, which will be described later, is predetermined information such as a device name or an authentication ID, and is an image that can be read from the information terminal **1** side. In particular, the authentication ID such as a passkey corresponding to the wireless communication interface is displayed on the display screen **20** as a coded image.

[0037] The user looks at the authentication information of the display screen **20** through the information terminal **1**. The information terminal **1** reads the authentication information of the display screen **20**, and executes device check as for whether wireless connection between own device and the display device **2** is appropriate or not by using the device name (device ID information) contained in the authentication information. The information terminal **1** obtains the authentication ID by decoding the coded image associated with the own device name. The information terminal **1** establishes wireless connection with the display device **2** using the authentication ID.

[0038] Note that an operation after the wireless connection between the information terminal **1** and the display device **2** is not limited. The information terminal **1** communicates with the display device **2**, and can thereby use functions such as the display function of the display device **2**. For example, the user wirelessly connects the information terminal **1** to a desired display device **2** in a store, and can then browse a moving image or the like displayed on the display screen **20** of the display device **2**. In the opposite way of thinking, it can be said that the display device **2** can use functions of the information terminal **1** thus wirelessly connected. Further, after use by the user, the wireless connection between the information terminal **1** and the display device **2** is cancelled. Further, in a case where the information terminal **1** is, for example, the HMD **1A**, it is possible to display an image of augmented reality (AR) or the like on a display surface **11** so as to match the display of the display screen **20** of the display device **2**.

[Wireless Connection Method]

[0039] FIG. **2** illustrates an outline of a wireless connection method according to the first embodiment. An upper side of FIG. **2** illustrates, as an example of the environment, a case of a 3 to 3 relationship in which three information terminals **1** and three display devices **2** exist. For example, it is assumed that a user **U1** wants to wirelessly connect his or her information terminal **T1** to a desired display device **D1**. It is also assumed that a user **U2** wants to wirelessly connect his or her information terminal **T2** to a desired display device **D3**. It is also assumed that a user **U3** just passes by, and does not carry out wireless connection with any display device **2**.

[0040] Note that as an application, it is not a configuration in which unlimited wireless connection is permitted for the information terminals **1** and the display devices **2** of all users, but a configuration is also possible that only wireless connection between an information terminal **1** of a specific user and a specific display device **2** is permitted. This configuration can be controlled through settings of authentication. For example, a control in which the information terminal **T1** of the user **U1** and the information terminal **T2** of the user **U2** cannot be wirelessly connected to a display device **D2** and other types of information terminals **1** of other users can be wirelessly connected to the display device **D2** is possible.

[0041] A procedure outline in the wireless connection method according to the first embodiment is as follows. At Step **S1**, a display device **2** detects information terminals **1** and users, which exist in the vicinity of the own device in environment by arbitrary detecting means. For example, the display device **2** detects information terminals **1** and users, which exist within a range in front of the camera **22**. More specifically, for example, the display device **2** may detect a state where there is a user who wears the HMD **1A** in front of the own device. For example, it is assumed that each of the display device **D1** and the display device **D3** detects the information terminal **T1** and the

information terminal T2. Further, at this time, the display device 2 detects information terminals 1 (in other words, wireless devices), each of which has a function capable of wireless connection, and obtains information such as a device name the detected information terminal 1.

[0042] At Step S2, the display device 2 displays information on the detected information terminals 1, in other words, the wireless devices, each of which becomes a wireless connection candidate, on display the screen 20. Specifically, the display device 2 displays a list containing information on each of the one or more information terminals 1 detected as described above on the display screen 20 as a device list. Although details will be described later, display information at this time contains a device name or a device address, for example. For example, the display device D1 displays information on each of the information terminal T1 and the information terminal T2 thus detected. The display device D3 also executes the similar operation.

[0043] At Step S3, the user visually recognizes and confirms the information on the list of the display screens 20 of the display devices 2, and carries out an operation of selecting an item corresponding to the device name of the own information terminal 1 from the list.

[0044] At Step S4, the display device 2 displays an image (coded image) in which an authentication ID regarding the selected information terminal 1 is coded on the display screen 20. Although details will be described later, the coded image displayed at this time is an image in which the selected device name and a passkey generated by the display device 2 side are coded, for example. For example, a QR (Quick Response) (registered trademark) code can be applied to the code, but the code is not limited to this.

[0045] At Step S5, the information terminal 1 reads the coded image displayed on the display screen 20 of the selected display device 2 desired by the user by the camera 12, and obtains authentication information. Specifically, the information terminal 1 extracts the device name and the passkey from the above coded image by decoding. Note that at this time, the user may direct a direction of the camera 12 of the information terminal 1 (for example, the HMD 1A) to the display screen 20 of the display device 2, and does not need to carry out an input operation of the authentication ID or the like. For example, the information terminal T1 of the user U1 recognizes display information of the display device D1. Similarly, the information terminal T2 of the user U2 recognizes display information of the display device D3.

[0046] Further, the information terminal 1 executes device check processing of whether the device name obtained by the above extraction matches a device name of the own device or not. In case of the matching, it means that it has been confirmed that the wireless connection in a one-to-one relationship between this information terminal 1 and the display device 2 of the other party is establishes appropriate. The information terminal 1 sets the above passkey to the own device after confirming the appropriateness by the device check.

[0047] At Step S6, the information terminal 1 uses the authentication information (in particular, device ID information) to confirm appropriateness of the wireless connection between the own device and the display device 2 of the other party (device check, which will be described later). After the confirmation, the information terminal 1 establishes the wireless connection between the own device and the display device 2 by using the passkey, and the display device 2 establishes wireless connection with the information terminal 1 by using the passkey.

[Information Terminal (HMD)—Appearance]

[0048] FIG. 3 illustrates an example of appearance configuration in a case where the information terminal 1 is the HMD 1A. The HMD 1A includes the display surfaces 11, the camera 12, a distance measuring sensor 14, sensor units 15, microphones 16, speakers 17, and the like on a spectacle-shaped chassis 10. The display surface 11 is a transmissive display surface corresponding to positions of both eyes. The camera 12 includes a right side camera 12R and a left side camera 12L, for example. Further, as illustrated in FIG. 3, the HMD 1A may be attached with an operation unit 19 such as a remote controller as input means. The operation unit 19 executes wireless communication with a body of the HMD 1A.

[Information Terminal (HMD)—Functional Block]

[0049] FIG. 4 illustrates an example of a functional block configuration in a case where the information terminal **1** is the HMD **1A**. The HMD **1A**, which is the information terminal **1**, includes a processor **101**, a memory **102**, the camera **12**, the distance measuring sensor **14**, a sensor unit **15**, a display unit **150**, a communication unit **160**, a voice input device **181**, a voice output device **182**, an operation input unit **183**, a battery **184**, and the like, and they are connected to each other by a bus or the like. The processor **101** is composed of a CPU, a ROM, a RAM, and the like, and corresponds to a controller that controls the entire HMD **1A**. The memory **102** is composed of a RAM, a storage, or the like, and stores therein information and data such as a control program **131**, an application program **132**, setting information **133**, display information **134**, and authentication information **135**. The camera **12** photographs an image of the outside world, and obtains image data. The distance measuring sensor **14** measures a distance to an object in the outside world. The sensor unit **15** includes various kinds of sensors including an acceleration sensor **141**, a gyro sensor **142**, a geomagnetism sensor **143**, a GPS receiver **144**, for example. The display unit **150** is a display device that includes the display surface **11**, a display driving circuit, an optical system, and the like. The communication unit **160** includes a communication device corresponding to various kinds of communication interfaces. Specifically, the communication unit **160** includes a short-distance communication unit and a long-distance communication unit. In the present embodiment, the short-distance communication unit corresponds to Bluetooth as a wireless communication interface. The long-distance communication unit corresponds to wireless communication interfaces such as a mobile network and a wireless LAN, for example. The voice input device **181** includes the microphone **16** described above. The voice output device **182** includes the speaker **17** described above. The operation input unit **183** includes buttons provided on the chassis **10**. Alternatively, the operation input unit **183** may be a portion that includes a wireless communication interface with the operation unit **19**. The battery **184** supplies electric power to each unit.

[0050] The information terminal **1** realizes each function by the processor **101** executing processes on the basis of the control program **131** and the like. Each function includes a function of executing processes in a flow, which will be described later. Each function includes functions such as device check, camera image analysis, decoding of a QR code, which will be described later.

[0051] The setting information **133** is system setting information or user setting information related to the functions. The display information **134** is data for being displayed on the display surface **11**. The authentication information **135** is information such as device ID information of the own device (a device name and a device address) and the authentication ID (the passkey) obtained from the display device **2**.

[Display Device—Functional Block]

[0052] FIG. 5 illustrates an example of a functional block configuration of the display device **2**. The display device **2** includes a processor **201**, a memory **202**, the camera **22**, a distance measuring sensor **24**, a sensor unit **25**, a display unit **250**, a communication unit **260**, a voice input device **281**, a voice output device **282**, an operation input unit **283**, a battery **284**, and the like, they are connected to each other by a bus or the like. The processor **201** is composed of a CPU, a ROM, a RAM, and the like, and corresponds to a controller that controls the entire display device **2**. The memory **202** is composed of a RAM, a storage, or the like, and stores therein information and data such as a control program **231**, an application program **232**, setting information **233**, display information **234**, and authentication information **235**. The camera **22** photographs an image of the outside world, and obtains image data. The distance measuring sensor **24** measures a distance to an object in the outside world. The sensor unit **25** includes various kinds of sensors including an acceleration sensor **241**, a gyro sensor **242**, a geomagnetism sensor **243**, and a GPS receiver **244**, for example. The sensor unit **25** may include a proximity sensor or the like. The display unit **250** is a display device that includes the display screen **20**, a display driving circuit, and the like, for

example, a touch panel. The communication unit **260** includes a communication device corresponding to various kinds of communication interfaces. Specifically, the communication unit **260** includes a short-distance communication unit and a long-distance communication unit. The short-distance communication unit corresponds to Bluetooth, for example. The long-distance communication unit corresponds to wireless communication interfaces such as a mobile network and a wireless LAN, for example. The voice input device **281** includes a microphone. The voice output device **282** includes a speaker. The operation input unit **283** includes buttons provided on a chassis. The operation input unit **283** may include a keyboard or the like. In that case, it becomes easy for the user to input characters. The battery **284** supplies electric power to each unit. The display device **2** may also be attached with an operation unit such as a remote controller.

[0053] The display device **2** realizes each function by the processor **201** executing processes on the basis of the control program **231** and the like. Each function includes a function of executing processes in a flow, which will be described later. Each function includes functions such as device check, user detection, device detection, passkey generation, and QR code generation, which will be described later.

[0054] The setting information **233** is system setting information or user setting information related to the functions. The display information **234** is data for being displayed on the display screen **20**. The authentication information **235** is information as such device ID information detected from the information terminal **1** and the generated authentication ID (the passkey).

[Wireless Communication Interface]

[0055] In the first embodiment, Bluetooth is applied as a wireless communication interface for wireless connection between the information terminal **1** and the display device **2**. In this case, a passkey in the Bluetooth standard is used as the authentication ID. Note that Wi-Fi may be applied as the wireless communication interface. In this case, an SSID and a password in the Wi-Fi standard is used as the authentication ID. This wireless connection system can similarly apply an authentication ID in accordance with a wireless communication interface to be applied.

[Processing Flow (1)]

[0056] FIG. **6** illustrates a processing sequence including communication between the information terminal **1** (for example, the HMD **1A**) and the display device **2** and a processing flow of each device in the wireless connection system and the method thereof according to the first embodiment. FIG. **6** has a flow including Steps **S100**, **S101**, . . . , **S105** for the information terminal **1**, and a flow including Steps **S200**, **S201**, . . . , and **S211** for the display device **2**.

[0057] At Step **S200**, the display device **2** detects users who exist in the vicinity thereof by using the camera **22** or the sensor unit **25**. In a case where a user comes near the front of the display screen **20**, for example, the display device **2** detects it. Note that it is also possible to omit Step **S200**. In a case where the display device **2** detects the users (Y), the processing flow proceeds to Step **S201**. Note that this user detection may be detection of an operation of pressing a hardware button on the operation input unit **283** provided in the display device **2** by the user, an operation of touching the touch panel, which is the display unit **250** by the user, or an operation of pressing a specific software button (GUI parts) in the display screen **20** by the user.

[0058] At Step **S201**, the display device **2** detects information terminals **1**, each of which is a wireless device that exists in the vicinity, by using the communication unit **260** (FIG. **5**), the camera **22**, or the sensor unit **25**. This wireless device detection is to receive and detect a beacon (a radio signal) emitted by the information terminal **1**, for example. The beacon has information such as a device name or a device address of the information terminal **1** (for the sake of explanation, which may be described as “device ID information”). In the present embodiment, at Step **S201**, the display device **2** obtains information on a pair of a device name and a device address of each information terminal **1** on the basis of the beacon from the corresponding information terminal **1**. Note that as the other detecting means, the display device **2** may determine and detect wireless devices on the basis of analysis of an image photographed by the camera **22**.

[0059] At Step S202, the display device 2 displays the following device list on the display screen 20 by using the information detected and obtained at Step S201.

[Screen Example (1)]

[0060] FIG. 7 illustrates a screen example when the device list is displayed on the display screen 20 of the display device 2 at Step S202. This screen 700 is a device list screen of a wireless connection menu screen. This screen 700 includes “wireless connection” menu display 701, “devices are detected” message display 702, and “available devices” display 703. The “available devices” display 703 includes device list display. As an example, this device list includes items 711, 712, and 713 as three device information items corresponding to a case where information terminals 1 (for example, the information terminals T1, T2, and T3 illustrated in FIG. 1 and FIG. 2), which are three wireless devices, are detected in the vicinity of this display device 2. A device name 704 is displayed in each device information item. This device name 704 is information such as a character string in a format that allows the user to identify the information terminal 1. The device name 704 to be displayed in the device information item is not limited to this, and may be other information such as a device address so long as it is a format that allows the user to visually identify the information terminal 1.

[0061] For example, it is assumed that the device name 704 (“Phone_APL”) in the item 711 is a device name of the information terminal T2; the device name 704 (“MAX-HMD01”) in the item 712 is a device name of the information terminal T1; and the device name 704 (“AR glass lite”) of the item 713 is a device name of the information terminal T3. For the sake of explanation, in FIG. 7, a code “(T1)” of the corresponding information terminal 1 is described in the device information item so that the correspondence relationship can be easily understood, but it is not actually displayed.

[0062] Each device information item is composed of GUI parts such as buttons that accept selection operations (for example, touch operations) by the user as well as the display of the device name 704. Alternatively, another selection button or the like may be provided in parallel with the device information item.

[0063] Note that other GUIs may be displayed in this screen 700. For example, a message image such as “Please select a device to be connected” may be displayed as information for guiding the user in his or her operation in an easy-to-understand manner. Further, a user interface is not limited to the display on the display screen 20, and voice output or the like may be used in combination.

[0064] Note that the three device (the information terminals 1) whose device information items are displayed in this device list screen are devices for which wireless connection is permitted. In case of devices for which wireless connection is not permitted, such items are not displayed. As an application, in case of setting to permit wireless connection between a specific information terminal 1 and a specific display device 2, the specific display device 2 displays a corresponding device information item only when the detected device corresponds to the specific information terminal 1.

[0065] Further, as another application, the display device 2 may be set with limitations such as the number of devices for simultaneous wireless connection. In that case, the display device 2 may display device information items so as to fall within the number of devices for simultaneous wireless connection. Alternatively, for example, in a case where the number of devices for simultaneous wireless connection is one, the display device 2 may display a plurality of device information items on the screen, and permit wireless connection to only one information terminal 1 of them by a method such as a first-come-first-served method. The applied control as described above can be realized additionally as a control of a layer different from the authentication of the present system.

[0066] Note that when the display device 2 displays a plurality of device information items in the device list screen or the like, the display device 2 may display a plurality of items side by side in accordance with the order of the detected wireless devices. Alternatively, the display device 2 may display a plurality of items side by side in accordance with the order of a distance of each of the

detected wireless devices. For example, the display device **2** may display a device information item of a wireless device estimated to exist in the nearest vicinity (that is, be likely to be selected) at the top thereof, or may display it in the most emphasized and enlarged manner.

[0067] The explanation returns to FIG. **6**. At Step **S203**, the display device **2** receives a selection operation of a device information item (the device name **704**) by the user from the device list. For example, the user **U1** recognizes the second item **712** having the device name **704** corresponding to the information terminal **T1**, which is the own device, from the device list of the screen **700** illustrated in FIG. **7**, and carries out a selection operation for it. The selection operation is a touch operation against the touch panel, which is the display unit **250** (FIG. **5**), for example. The display device **2** grasps the information terminal **1** (for example, the information terminal **T1**) having the device name **704**, which is selected by the user.

[0068] At Step **S204**, the display device **2** generates a passkey, which is an authentication ID for wireless connection, for the above selected device (the information terminal **1**).

[0069] At Step **S205**, the display device **2** generates a predetermined code (coded image) according to the first embodiment by using a set of the device name selected at Step **S203** described above and the passkey generated at Step **S204** described above. This code is generated in the form of a QR code, which is a two-dimensional code, for example.

[0070] At Step **S206**, the display device **2** displays authentication information, which contains the coded image generated at Step **S205** described above, in the display screen **20**.

[Screen Example (2)]

[0071] FIG. **8** illustrates a screen example when a coded image is displayed on the display screen **20** of the display device **2** at Step **S206**. This screen **800** is a code screen of the wireless connection menu screen. This screen **800** includes “please read code” message display **802**, and authentication information display **803**. The authentication information display **803** is one item, and includes a pair of a device name **804** and a code (coded image) **805**. The device name **804** has the same value as the device name **704** of the selected item **712** illustrated in FIG. **7**. The device name **804** is information supplementarily provided so that the user can easily grasp the items of this authentication information display **803** when the user views it, that is, the user can grasp that it is an item regarding the own device. The display of the device name **804** can be omitted. The code **805** is a QR code, and must be displayed. This QR code includes the selected device name, and is one of features. The user looks at the authentication information display **803** on the screen **800**.

[0072] The explanation returns to FIG. **6**. On the other hand, at Step **S100**, the information terminal **1** causes the camera **12** to read the code **805** displayed on the display screen **20** as illustrated in FIG. **8** on the basis of an operation of the user, for example, an operation of causing the camera **12** of the HMD **1A** to photograph the display screen **20**. The information terminal **1** recognizes a portion of the code **805** from an image photographed by the camera **12**, and extracts the set of the device name and the passkey from the code **805** by decoding.

[0073] At Step **S101**, the information terminal **1** executes device check processing on the information terminal **1** side by using the set of the device name and the passkey obtained by extraction at Step **S100** described above. This device check processing is confirmation of whether the device name obtained from the code **805** matches the device name of the own device or not. In other word, this device check processing is confirmation of whether one-to-one wireless connection between this information terminal **1** and the display device **2** in front of the user is appropriate or not. The device check at Step **S101** is essential in the sequence according to the first embodiment, and is one of the features. In a case where the device names match each other (OK), the processing flow proceeds to Step **S102**. In a case where the device names do not match each other (NG), the processing flow proceeds to Step **S104**. In a case where the device names match each other, it represents that the information terminal **1** and the display device **2** of the other party have a relationship of appropriate wireless connection. In a case where the device names do not match each other, it represents that the information terminal **1** and the display device **2** of the other party

do not have a relationship of appropriate wireless connection, that is, that wireless connection is not permitted.

[0074] At Step S104, the information terminal 1 executes error display on the display surface 11 as error processing. As a result, the flow on the information terminal 1 side is ended. In this case, the flow restarts from the beginning thereof.

[0075] At Step S102, the information terminal 1 sets the passkey obtained at Step S100 to the circuit of the communication unit 160 (FIG. 4) as the authentication ID for the wireless connection.

[0076] At Step S103, the information terminal 1 transmits a wireless connection request to the display device 2 by communication processing of the communication unit 160. At this time, a circuit of the communication unit 160 executes the communication processing using the passkey set at Step S102. In a communication frame of this wireless connection request, information containing the passkey is described together with the address and the device name of a transmission source (the information terminal 1).

[0077] On the other hand, at Step S207, the display device 2 receives the wireless connection request from the information terminal 1 by the communication processing of the communication unit 260 (FIG. 5). At this time, a circuit of the communication unit 260 executes communication processing by using the passkey generated at Step S204. The display device 2 takes out the passkey from the transmission source (the information terminal 1) in the wireless connection request, and receives the wireless connection request as a proper request in a case where the passkey matches the passkey at Step S204.

[0078] At Step S208, the display device 2 executes device check processing on the display device 2 side by using the device name taken out from the request of the information terminal 1 at Step S207 and the device name generated at Step S205. In this device check processing, for example, the display device 2 confirms whether the device name at Step S207 matches the device name at Step S204 or not. In a case where the device names match each other (OK), the processing flow proceeds to Step S209. In a case where the device names do not match each other (NG), the processing flow proceeds to Step S211. In a case where the device names match each other, it represents that wireless connection between the information terminal 1 of a request source and this display device 2 is appropriate. In a case where the device names do not match each other, it represents that the wireless connection between the information terminal 1 of the request source and this display device 2 is not appropriate, that is, that the wireless connection is not permitted.

[0079] At Step S211, the display device 2 executes error display on the display screen 20 as error processing. As a result, the flow on the display device 2 side is ended. In this case, the flow restarts from the beginning thereof.

[0080] At Step S209, the display device 2 sets the passkey generated at Step S204 described above to the circuit of the communication unit 260 (FIG. 5) as the authentication ID for the wireless connection. As a result, the same passkey is set to both the information terminal 1 and the display device 2.

[0081] At Step S210, the display device 2 establishes the wireless connection with the information terminal 1 by the communication processing of the communication unit 260. Correspondingly, at Step S105, the information terminal 1 establishes the wireless connection with the display device 2 by the communication processing of the communication unit 160. After Steps S105 and S210, the present sequence is ended. In a state where the wireless connection between the information terminal 1 and the display device 2 has been established, it is possible to actually transmit and receive data between the devices. Then, the wireless connection between these devices is disconnected if necessary.

[0082] Note that a characteristic portion (a portion related to main authentication) in the sequence according to the first embodiment is shown by a broken line frame. In particular, the device check at Step S101 is a unique process using the device name, and is a process of confirming an appropriate relationship in a case where the display device 2 is viewed from the information

terminal **1**. It is also a process of confirming an appropriate one-to-one relationship between the information terminal **1** and the display device **2** even in multiple-to-multiple environment. Further, in particular, this sequence also includes the device check on the display device **2** side at Step **S208** in addition to the device check on the information terminal **1** side at Step **S101**. This process is also a unique process using the device name, and is a process of confirming an appropriate relationship in a case where the information terminal **1** is viewed from the display device **2**. In the first embodiment, simple and appropriate wireless connection that does not require the user to carry out an input operation for a recognition ID can be realized by the sequence including the mutual device checks at Step **S101** and Step **S208**.

[0083] Note that as a modification example of the first embodiment, it is also possible to omit the device check processing on the display device **2** side at Step **S208**.

[Effects and The Like (1)]

[0084] As described above, according to the first embodiment, it is possible to reduce the time and effort of the user to carry out the input operation for the authentication ID such as a passkey when the information terminal **1** of the user is wirelessly connected to the display device **2** in the environment. Even in the multiple-to-multiple environment, appropriately secure wireless connection *n* between the desired information terminal **1** thus selected and the display device **2**. The user may carry out an operation of selecting an item corresponding to the own device on the display screen **20** (FIG. 7) of the display device **2**, and causing the information terminal **1** to read a displayed code (FIG. 8). Then, the information terminal **1** automatically executes the device check (**S101** of FIG. 6), and confirms the appropriateness of the wireless connection between the own device and the display device **2** of the other party. As a result, the passkey is set automatically (**S102**, **S209**), and the wireless connection is established. Thus, the user can quickly make wireless connection between the desired information terminal **1** and the desired display device **2** with little effort and without mistakes even in multiple-to-multiple environment.

[0085] The case where the wireless connection is permitted in a basically free combination between the plurality of devices and the plurality of devices in the environment has been described in the first embodiment. However, the present invention is not limited to this. As an application, more detailed control regarding availability of wireless connection is also possible. For example, settings regarding authentication can be made in advance so that wireless connection of a certain type of information terminal **1** of a user is permitted only with a certain type of display device **2**. This authentication can be executed as authentication of a different layer from that of the authentication (including the device check) that has been described in the first embodiment and the like.

[0086] As an example of usage, in a store such as a restaurant, each user who is a customer can cause his or her information terminal **1** to be wirelessly connected to a desired display device **2** selected from a plurality of display devices **2** (for example, tablet terminals) respectively installed on tables or the like in the store easily, and use a display function, an order function, and a payment function included in the display device **2**. As another example, payment devices each having a cashless payment function are prepared as a plurality of display devices **2** in a sales store or the like. A customer can cause his or her information terminal **1** to be wirelessly connected to a selected payment device, and use the cashless payment function thereof. As still another example, a display device **2** is installed for each seat in a vehicle. A passenger can cause his or her information terminal **1** to be wirelessly connected to a selected the display device **2**, and use a display function for a moving image or the like, and an Internet connection function. Further, in any of the examples, it is not limited to the use of an unspecified number of people, it is also possible to restrict the use so that only wireless connection between an information terminal **1** of a specific user and a specific display device **2** and the use of functions are permitted through settings of authentication.

[0087] In the above, the case where the user uses the HMD **1A** as the information terminal **1** has mainly been described. However, the present invention can similarly be applied to a case where the

smartphone 1B (FIG. 1) is used. In this case, the user operates the smartphone 1B held by his or her hand. For example, the user may cause a camera of the smartphone 1B to photograph the display screen 20 of the display device 2.

Second Embodiment

[0088] A wireless connection system and a method thereof according to a second embodiment of the present invention will be described with reference to FIG. 9 to FIG. 11. Hereinafter, portions of the second embodiment and the like different from those of the first embodiment will be described. A system configuration of the second embodiment and the like is similar to that illustrated in FIG. 1 and the like. As a different configuration point, there is a processing sequence including authentication. In the second embodiment, an AR image display function by the HMD 1A is used.

[Processing Flow (2)]

[0089] FIG. 9 illustrates a processing sequence including communication between an information terminal 1 (for example, an HMD 1A) and a display device 2 and a processing flow of each device in a wireless connection system and a method thereof according to the second embodiment. FIG. 9 includes Steps S121 to S129 for the information terminal 1 and Steps S220 to S229, S229A, and S229B for the display device 2. In FIG. 9, Steps mainly different from those in FIG. 6 according to the first embodiment are Steps S121, S122, and S123, and Steps S221, S222, and S223. Steps after Step S124 and Steps after Step S224 are substantially similar to Steps after Step S100 and Steps after Step S204 of FIG. 6.

[0090] At Step S220, the display device 2 detects users who exist in the vicinity thereof, and detects information terminals 1, which are wireless devices that exist in the vicinity thereof. After Step S221, at Step S222, the display device 2 creates a device list including codes unique in the second embodiment (referred to as a “first code”) by using information such as device names detected and obtained as described above. In the second embodiment, this first code is a coded image such as a QR code in which a device name is coded for each detected the information terminal 1, and does not include the passkey described above. Information to be coded is not limited to the device name, and may be another information such as a device address.

[0091] At Step S202, the display device 2 displays the device list created as described above on the display screen 20 as described below.

[Screen Example (3)]

[0092] FIG. 10 illustrates a screen example when the device list including the first code is displayed on the display screen 20 of the display device 2 at Step S222. This screen 1000 is a device list display screen of a wireless connection menu screen. This screen 1000 includes “devices are detected” message display 1002 and device list display 1003. Device information items for each detected device (the information terminal 1 are displayed in parallel in the device list display 1003 as the device list. In the present embodiment, items 1011, 1012, and 1013 are included as three device information items regarding the three detected information terminals 1. Each device information item includes device name display 1004 and QR code display 1005. The device name display 1004 is auxiliary information for conveying the detected devices to a user. A QR code of the QR code display 1005 is the first code.

[0093] The explanation returns to FIG. 9. At Step S223, the user carries out a selection operation for a device information item corresponding to his or her information terminal 1 (the HMD 1A), for example, the item 1012 from the device list of this screen 1000 with reference to the device name display 1004. At this time, in the second embodiment, Steps S121 to S123 are provided to support this selection operation.

[0094] At Step S121, the information terminal 1 reads the device list (FIG. 10) of the display screen 20 of the display device 2 by the camera 12. The information terminal 1 recognizes the device name display 1004 or the QR code display 1005 of each device information item in the device list display 1003 from an image of the camera 12.

[0095] At Step S122, the information terminal 1 confirms whether there is the device name display

1004 corresponding to the device name (character string) of the own device (the HMD **1A**) or not from the above read device list. In a case where there is one, the information terminal **1** detects it. Alternatively, the information terminal **1** may execute similar determination and detection by using the extracted device name by decoding the QR code of the QR code display **1005**. In the example of FIG. **10**, it is assumed that the HMD **1A** (the information terminal **T1** illustrated in FIG. **1**) detects the device name (“MAX-HMD01”) of the device name display **1004** in the item **1012**. In a case where there is an own device name (Y), the processing flow proceeds to Step **S123**. In a case where there is no own device name (N), the processing flow proceeds to Step **S128**, and error display is executed.

[0096] At Step **S123**, the information terminal **1** displays, on the display surface **11**, an image for conveying the device information item corresponding to the detected own device name to the user. This image is an AR image for highlighting the item and prompting the user to operate it. This AR image is superimposed and displayed on the device information item (the item **1012**) of the display screen **20** as described below.

[Screen Example (4)]

[0097] FIG. **11** illustrates an example of displaying the image at Step **S123** on the display surface **11** of the HMD **1A**. (A) of FIG. **11** illustrates a schematic image when a user (whose illustration is omitted) looks at the display screen **20** of the display device **2** (for example, the screen **1000** illustrated in FIG. **10**) through the HMD **1A** worn on a head of the user. (B) of FIG. **11** illustrates a schematic image when the item **1012** of the screen **1000** is viewed from a point of view of the user through the display surface **11** of the HMD **1A**. The display surface **11** is a transmissive display surface capable of displaying an AR image. The camera **12** of the information terminal **1** captures the display screen **20**. The user looks at the device list of the display screen **20**. Here, the display device **2** that can be seen through the display surface **11** is schematically illustrated by a broken line. The HMD **1A** (the display unit **150** illustrated in FIG. **4**) displays an AR image **1101** on the display surface **11** on the basis of analysis and the like of the image of the camera **12** so as to be superimposed and displayed on the item **1012** (the item corresponding to an own device name) of the screen **1000**. This AR image **1101** is an image having a meaning of prompting the user to carry out a selection operation for this device information item. In the present embodiment, the AR image **1101** is a frame image that surrounds the item **1012**, but is not limited to this. For example, the AR image **1101** may be an image that surrounds the device name display **1004** illustrated in FIG. **10**, an image that surrounds the QR code display **1005**, or an image such as an arrow accompanied by a message such as “Please select this item”. The user can understand that the user may pay attention to the item **1012** to carry out a selection operation in accordance with this AR image **1101**. This makes the selection operation at Step **S223** easier.

[0098] The explanation returns to FIG. **9**. At Step **S203**, the user carries out a selection operation for an item corresponding to the own device from the above device list in accordance with the AR image **1101** at Step **S123**. The display device **2** receives the selection operation, and grasps the selected device name.

[0099] Steps after Step **S224** are substantially similar to Steps after Step **S204** illustrated in FIG. **6** according to the first embodiment. At Step **S224**, the display device **2** generates a passkey for the above selected device. At Step **S225**, the display device **2** uses the above passkey and the above selected device name to generate a predetermined code as a second code according to the second embodiment in the format of a QR code, for example. At Step **S206**, as well as FIG. **8**, the display device **2** displays the above generated code (the second code) on the display screen **20**. On the other hand, at Step **S124**, the information terminal **1** reads the code (the second code) on the display screen **20** on the basis of an operation of the user, and extracts a set of the device name and the passkey by decoding. At Step **S125**, the information terminal **1** executes device check processing by using the above set of the device name and the passkey. In a case where the extracted device name and the own device name match each other (OK), the processing flow proceeds to

Step **S126**. In a case where the device names do not match each other (NG), the processing flow proceeds to Step **S128**, and error display is executed. The device check at Step **S125** is an essential process in the second embodiment.

[0100] At Step **S126**, the information terminal **1** sets the passkey obtained at Step **S124** to a circuit of the communication unit **160** (FIG. **4**). At Step **S127**, the information terminal **1** transmits a wireless connection request to the display device **2** by communication processing of the communication unit **160**. On the other hand, at Step **S227**, the display device **2** receives the wireless connection request from the information terminal **1** by communication processing of the communication unit **260** (FIG. **5**). The display device **2** takes out information such as the passkey from the received wireless connection request. At Step **S228**, the display device **2** executes device check processing by using the information such as the passkey obtained at Step **S227**. In a case where device names match each other (OK), the processing flow proceeds to Step **S229**. In a case where the device names do not match each other (NG), the processing flow proceeds to Step **S229B**, and error processing is executed. At Step **S229**, the display device **2** sets the above passkey to a circuit of the communication unit **260**. At Step **S229A**, the display device **2** establishes wireless connection with the information terminal **1** by the communication processing of the communication unit **260**. Correspondingly, at Step **S129**, the information terminal **1** establishes wireless connection with the display device **2** by the communication processing of the communication unit **160**.

[Effects and The Like (2)]

[0101] As described above, in the second embodiment, the information terminal **1** of the user automatically recognizes the item corresponding to the own device from the device list of the display screen **20** of the display device **2**, and conveys it to the user by the AR image (Step **S123** of FIG. **9**, FIG. **11**). As a result, the user can easily select the item corresponding to the own device, and quickly proceed to the procedure for authentication to make wireless connection.

[0102] Note that as a modification example of the second embodiment, it is also possible to omit the device check processing on the display device **2** side.

Third Embodiment

[0103] A wireless connection system and a method thereof according to a third embodiment of the present invention will be described with reference to FIG. **12** and FIG. **13**. In the third embodiment, an information terminal **1** finds an own device from a device list on a display screen **20** of a display device **2**, and automatically executes authentication and wireless connection without requiring a selection operation by a user.

[Processing Flow (3)]

[0104] FIG. **12** illustrates a processing sequence including communication between an information terminal **1** (for example, an HMD **1A**) and a display device **2** and a processing flow of each device in a wireless connection system and a method thereof according to the third embodiment. FIG. **12** includes Steps **S131** to **S136** for the information terminal **1** and Steps **S230** to **S238** for the display device **2**. As a configuration point mainly different from FIG. **6** and FIG. **9**, FIG. **12** includes Steps **S131** and **S132** for the information terminal **1**, and Step **S233** for the display device **2**.

[0105] At Step **S230**, the display device **2** executes user detection and device detection. At Step **S231**, the display device **2** generates a passkey. At Step **S232**, the display device **2** generates a predetermined code according to the third embodiment. This code is an image obtained by coding a set of a device name and the passkey in the format of a QR code, for example. Next, at Step **S233**, the display device **2** displays a device code list created using the code generated at Step **S232** on the display screen **20**.

[Screen Example (5)]

[0106] FIG. **13** illustrates a screen example when the device code list is displayed on the display screen **20** of the display device **2** at Step **S233**. This screen **1300** is a device code list display screen of a wireless connection menu screen. This screen **1300** includes “Devices are detected” message

display **1302**, and device code list display **1303**. In the device code list display **1303**, device code information items for each detected device (the information terminal **1**) are displayed in parallel as a device code list. In the present embodiment, items **1311**, **1312**, and **1313** are provided as three device code information items regarding three detected information terminals **1**. Each device code information item includes device name display **1304** and QR code display **1305**. The device name display **1304** is auxiliary information for conveying a detected device to the user, and display thereof can be omitted. A QR code of the QR code display **1305** is a predetermined code according to the third embodiment, and is obtained by coding the set of the device name and the passkey. It can be said that the screen **1300** illustrated in FIG. **13** is streamlined by integrating the device list screen illustrated in FIG. **7** and the code screen illustrated in FIG. **8** according to the first embodiment into one.

[0107] The explanation returns to FIG. **12**. At Step **S131**, the information terminal **1** reads the device code list on the display screen **20** of the display device **2** by a camera **12**. The information terminal **1** recognizes the device name display **1304** and the QR code display **1305** of each device code information item from an image of the camera **12**.

[0108] At Step **S132**, the information terminal **1** extracts the set of the device name and the passkey from the QR code of the QR code display **1305** obtained at Step **S131** by decoding. The information terminal **1** executes device check processing by using the extracted device name. This processing is a process of extracting, in a case where there is a device name of the own device (the HMD **1A**), the device name from one or more obtained device names, and is a unique and indispensable process in the third embodiment. In a case where the own device name can be detected (Y), the processing flow proceeds to Step **S133**. In a case where the own device name cannot be detected (N), the processing flow proceeds to Step **S135**, and error display is executed.

[0109] Processes after Step **S133** are substantially similar to those according to the first embodiment described above. At Step **S133**, the information terminal **1** sets the passkey obtained at Step **S131** to a circuit of the communication unit **160**. At Step **S134**, the information terminal **1** transmits a wireless connection request to the display device **2** by communication processing of the communication unit **160**. On the other hand, at Step **S234**, the display device **2** receives the wireless connection request from the information terminal **1** by communication processing of the communication unit **260**. The display device **2** takes out information such as the device name and the passkey from the received wireless connection request. At Step **S235**, the display device **2** executes device check processing on the display device **2** side by using the device name at Step **S232** and the device name obtained at Step **S234**. In a case where the device names match each other (OK), the processing flow proceeds to Step **S236**. In a case where the device names do not match each other (NG), the processing flow proceeds to Step **S238**, and error processing is executed. At Step **S236**, the display device **2** sets the above passkey to a circuit of the communication unit **260**. At Step **S237**, the display device **2** establishes wireless connection with the information terminal **1** by the communication processing of the communication unit **260**. Correspondingly, at Step **S136**, the information terminal **1** establishes wireless connection with the display device **2** by the communication processing of the communication unit **160**.

[Effects and The Like (3)]

[0110] As described above, in the third embodiment, the information terminal **1** of the user recognizes the item corresponding to the own device from the device code list (FIG. **13**) on the display screen **20** of the display device **2**, and automatically proceed to the procedure for authentication to make wireless connection quickly.

[0111] Note that as a modification example of the third embodiment, it is also possible to omit the device check processing on the display device **2** side.

Fourth Embodiment

[0112] A wireless connection system and a method thereof according to a fourth embodiment of the present invention will be described with reference to FIG. **14** and FIG. **15**. In the fourth

embodiment, temporary connection is first executed between an information terminal **1** and a display device **2**, and authentication and wireless connection are then executed.

[Processing Flow (4)]

[0113] FIG. **14** illustrates **1000**) a processing sequence including communication between an information terminal **1** (for example, an HMD **1A**) and a display device **2** and a processing flow of each device in a wireless connection system and a method thereof according to the fourth embodiment. FIG. **14** includes Steps **S141** to **S149** for the information terminal **1** and Steps **S240** to **S248** for the display device **2**. As a mainly different configuration point, FIG. **14** includes Steps **S141** to **S144** for the information terminal **1** and Steps **S241** to **S244** for the display device **2**.

[0114] At Step **S240**, the display device **2** executes user detection and device detection. At Step **S141**, the information terminal **1** sets a temporary passkey for temporary connection to a circuit of the communication unit **160** (FIG. **4**). For the temporary passkey, for example, if a passkey has 6 digits, a fixed value such as 0000000 is used. On the other hand, at Step **S241**, the display device **2** sets the same temporary passkey to a circuit of the communication unit **260** (FIG. **5**).

[0115] At Step **S142**, the information terminal **1** transmits a wireless connection request for the temporary connection to the display device **2** by using the above temporary passkey. At Step **S242**, the display device **2** receives the wireless connection request. As a result, at Steps **S143** and **S243**, temporary wireless connection is executed between the information terminal **1** and the display device **2**.

[0116] At Step **S144**, the information terminal **1** generates a passkey for wireless connection for a non-temporary purpose, and transmits it to the display device **2**. At Step **S244**, the display device **2** receives the passkey from the information terminal **1**.

[0117] At Step **S245**, the display device **2** generates a predetermined code according to the fourth embodiment. In the fourth embodiment, this code is an image in which only a device name is coded in the format of a QR code, for example. At Step **S246**, the display device **2** displays the above generated code on the display screen **20** as described below.

[Screen Example (6)]

[0118] FIG. **15** illustrates a screen example when the code is displayed on the display screen **20** of the display device **2** at Step **S246**. This screen **1500** is a code screen of a wireless connection menu screen. This screen **1500** includes “Please read code” message display **1502** and code display **1503**. The code display **1503** is a QR code related to a device name of one target device (the information terminal **1**).

[0119] The explanation returns to FIG. **14**. At Step **S145**, the information terminal **1** reads the code (the code display **1503**) displayed on the above display screen **20** by a camera **12**. The information terminal **1** extracts the device name from the code by decoding. Then, at Step **S146**, similar to the first embodiment, the information terminal **1** executes device check processing by using the above extracted device name. The information terminal **1** confirms whether the above extracted device name matches a device name of an own device (the HMD **1A**) or not. In a case where the device names match each other (OK), the processing flow proceeds to Step **S146**. In a case where the device names do not match each other (NG), the processing flow proceeds to Step **S149**, and error display is executed. At Step **S146**, the information terminal **1** sets the passkey generated at Step **S144** to the circuit of the communication unit **160** for the target wireless connection. On the other hand, at Step **S247**, the display device **2** side also sets the passkey received at Step **S244** to the circuit of the communication unit **260** for the target wireless connection. As a result, at Steps **S148** and **S248**, the wireless connection between the information terminal **1** and the display device **2** is established. In other word, the temporary wireless connection is ended, and the target wireless connection is achieved.

[Effects and The Like (4)]

[0120] As described above, in the fourth embodiment, after the information terminal **1** and the display device **2** are temporarily connected, the information terminal **1** can execute the device

check using the code of the device name, and quickly executes the wireless connection.

[0121] Note that as a modification example of the fourth embodiment, it is also possible to omit the device check processing on the display device 2 side.

Another Embodiment

[0122] The following is also possible as another embodiment. In the configuration illustrated in FIG. 1 according to the first embodiment, the case where the information terminal 1 and the display device 2 are wirelessly connected to each other directly at a short distance has been described.

However, the present invention is not limited to this. It is also possible to make wireless connection between the information terminal 1 and the display device 2 via a device such as a wireless base station or an access point is also possible. Further, it is also possible for the information terminal 1 and the display device 2 to communicate with and cooperate with an external apparatus such as a server on a communication network.

[0123] As described above, the present invention has specifically been described on the basis of the embodiments. However, the present invention is not limited to the embodiments described above, and the present invention may be modified into various forms without departing from the substance thereof.

REFERENCE SIGNS LIST

[0124] 1, T1, T2, T3, TM . . . information terminal, 1A . . . HMD, 1B . . . smartphone, 2, D1, D2, DN . . . display device, 11 . . . display surface, 12 . . . camera, 20 . . . display screen, 22 . . . camera, U1, U2, U3, UM . . . user.

Claims

1. A wireless connection system for executing wireless connection between a display device and a plurality of information terminals including a first information terminal, wherein the display device comprises a communication interface, an operation input interface, a display and a processor, wherein the processor is configured to: in response to receiving a wireless signal transmitted from the first information terminal by the communication interface, detect first device ID information for identifying the first information terminal included in the wireless signal, display a device list including a first device information item corresponding to the detected first device ID information, in response to accepting a selection operation for selecting the first device information item included in the device list by the operation input interface, code the first device ID information corresponding to the selected first device information item and first authentication ID information for the first information terminal that has the first device ID information, and display a first coded image that is coded the first device ID information and the first authentication ID information on the display, wherein the first information terminal comprises a first communication interface, a first memory, a first camera and a first processor, wherein the first processor is configured to: in response to recognizing the first coded image from a photographed image photographed by the first camera, extract a device ID information and an authentication ID information by decoding the first coded image, confirm whether the extracted device ID information matches the first device ID information stored in the first memory, and in response to matching the extracted device ID information with the first device ID information, transmit a connection request to the display device using the authentication ID information by the first communication interface.

2. The wireless connection system according to claim 1, wherein the first coded image is a two-dimensional code image.

3. The wireless connection system according to claim 1, wherein the plurality of information terminals includes a second information terminal, and wherein, in response to detecting second device ID information included in a wireless signal transmitted from the second information terminal after the first device ID information is detected, the processor of the display device is configured to display the first device information item and a second device information item

corresponding to the second device ID information in the device list.

4. The wireless connection system according to claim 3, wherein the first information terminal and the second information terminal are head mounted displays.

5. The wireless connection system according to claim 1, wherein the plurality of information terminals includes a second information terminal, wherein the display device further comprises a distance measuring sensor, and wherein, in response to detecting second device ID information included in a wireless signal transmitted from the second information terminal after the first device ID information is detected, the processor the display device is configured to display the first device information item and a second device information item corresponding to the second device ID information in the device list in accordance with an order of a distance of each of the first information terminal and the second information terminal.

6. The wireless connection system according to claim 1, wherein the processor of the display device is configured to display device information items in the device list so as to fall within the number of devices for simultaneous wireless connection.

7. The wireless connection system according to claim 1, wherein the processor of the display device is configured to display a message for prompting to select one of the device information items in the device list along with the device list on the display.

8. The wireless connection system according to claim 1, wherein the first device ID information is a device name or a device address of the first information terminal.

9. A wireless connection system for executing wireless connection between a display device and a plurality of information terminals including a first information terminal, wherein the display device comprises a communication interface, an operation input interface, a display and a processor, wherein the processor is configured to: in response to receiving a wireless signal transmitted from the first information terminal by the communication interface, detect first device ID information for identifying the first information terminal included in the wireless signal, and display a device list including a first device information item corresponding to the detected first device ID information, wherein the first information terminal comprises a first communication interface, a first memory, a first camera, a first display and a first processor, wherein the first processor is configured to: obtain a photographed image of the device list displayed on the display of the display device by the first camera, detect whether the device list includes the first device information item corresponding to the first device ID information stored in the memory based on the photographed image, and in response to detecting that the device list includes the first device information item, display a predetermined image on the first display so as to superimpose on the first device information item in the device list, wherein the processor of the display device is further configured to: in response to accepting a selection operation for selecting the first device information item included in the device list by the operation input interface, code the first device ID information corresponding to the selected first device information item and first authentication ID information for the first information terminal that has the first device ID information, and display a first coded image that is coded the first device ID information and the first authentication ID information on the display, wherein the first processor of the first information terminal is further configured to: in response to recognizing the first coded image from a photographed image photographed by the first camera, extract a device ID information and an authentication ID information by decoding the first coded image, confirm whether the extracted device ID information matches the first device ID information stored in the first memory, and in response to matching the extracted device ID information with the first device ID information, transmit a connection request to the display device using the authentication ID information by the first communication interface.

10. The wireless connection system according to claim 9, wherein the predetermined image is an image for prompting a user of the first information terminal to carry out a selection operation for the first device information item.

11. The wireless connection system according to claim 9, wherein the predetermined image is an

augmented reality image.

12. The wireless connection system according to claim 11, wherein the first information terminal is a head mounted display.

13. The wireless connection system according to claim 9, wherein the plurality of information terminals includes a second information terminal, and wherein, in response to detecting second device ID information included in a wireless signal transmitted from the second information terminal after the first device ID information is detected, the processor of the display device is configured to display the first device information item and a second device information item corresponding to the second device ID information in the device list.

14. The wireless connection system according to claim 13, wherein the first information terminal and the second information terminal are head mounted displays.

15. The wireless connection system according to claim 9, wherein the display device further comprises a distance measuring sensor, wherein the plurality of information terminals includes a second information terminal, and wherein, in response to detecting second device ID information included in a wireless signal transmitted from the second information terminal after the first device ID information is detected, the processor of the display device is configured to display the first device information item and a second device information item corresponding to the second device ID information in the device list in accordance with an order of a distance of each of the first information terminal and the second information terminal.

16. The wireless connection system according to claim 9, wherein the processor of the display device is configured to display device information items in the device list so as to fall within the number of devices for simultaneous wireless connection.

17. The wireless connection system according to claim 9, wherein the first device ID information is a device name or a device address of the first information terminal.
